

Brady Garrison

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EDUCATION

Brown University

Providence, RI | Expected Graduation May 2026

Concentrations: *ScB Mechanical Engineering (Honors Eligible)* and *AB Computer Science*; Current GPA: 3.75/4.00

- **Relevant Coursework:** Embedded Systems, Fluid Dynamics, Solid Mechanics

Pembroke Pines Charter High School

Pembroke Pines, FL | Class of 2022

Unweighted GPA: 4.00/4.00; Weighted GPA: 5.62/4.00; Class Rank: 1/452; **SAT: 1590** (800 math, 790 reading)

PROFESSIONAL EXPERIENCE

Tiffany & Co.

Cumberland, RI | June 2024 – August 2024

Manufacturing Process Engineering Summer Intern

- Updated **30 work instruction applications** that are used throughout global Tiffany manufacturing facilities using JD Edwards and Tulip software.
- Created and tested applications for inventory management, machine maintenance, and evaluation of machine efficiency in the machine shop for Tiffany & Co.'s largest manufacturing facility. Applications **increased individual time efficiency by up to 50%** at key manufacturing bottlenecks.
- Engaged with **cross functional teams** of automation, industrial, and quality assurance engineers to streamline factory efficiency and better understand product pipelines.

Brown University, School of Engineering

Providence, RI | September 2023 – December 2023

Undergraduate Teaching Assistant

- **Developed three new laboratory projects for student groups** involving soft materials engineering, robotics, and the properties of shape memory alloys. This included the development of bills of materials, assembly procedures, programming frameworks, and rapid prototypes.
- Mentored **5+ student project groups each with 5-8 members** to implement project plans. Organized weekly meetings and status presentations. Taught design principles, manufacturing, and review processes critical for new engineers.

Naval Research Laboratory, Distributed Intelligent Systems

Washington, DC | June 2021 – July 2022

Science and Engineering Apprenticeship Program, High School Intern

- **Implemented Artificial Intelligence algorithms** in Python to compete in limited information chess competition.
- Presented my findings and methodology at the Neural Information Processing Systems conference. **Publication Co-Author.**

ENGINEERING AND DESIGN EXPERIENCE

Brown University Wilhelmus Lab

Providence, RI | September 2024 – Present

Undergraduate Research Assistant

- Developed **motion control programs** for a novel underwater robot using Arduino and Python.
- **Integrating flow and tactile sensors** into a whisker inspired design for environmental mapping.
- **Collaborating with a diverse team** of post-doctoral researchers and Ph.D. students specializing in fluid dynamics, computer science, and mechanical design to prototype autonomous underwater vehicle concepts.

Brown University Formula Racing Team

Providence, RI | September 2022 – December 2024

Brakes Subsystem Lead (2023), Steering Subsystem Lead (2024)

- Led design of steering subsystem that **decreased weight by approximately 25%** while improving responsiveness and ergonomics. Developed custom carbon fiber steering column and steering wheel using first principles analysis, computer aided design, and iterative testing
 - At 2024-2025 SAE IC Formula Competition **top 11 design** performance at competition, **top 18 overall team**.
- Spearheaded braking subsystem that exceeded safety standards and **received 4/5 design score** at 2023-2024 SAE IC Formula Competition. **Max braking temperature decrease of 64%**.
- **Mentored 5+ new members** each year, integrating them into important projects within the driver controls subsystem. Promoted development of technical skills and understanding of engineering design process on a large team.

TECHNICAL SKILLS / PERSONAL INTERESTS

Software: SolidWorks, MATLAB, ANSYS, Tulip, Fusion 360, Git, Arduino, Blender, and Microsoft Office

Programming Languages: Python (TensorFlow, NumPy, Matplotlib, Scikit), Java, JavaScript, C, and HTML

Interests: Running, Snorkeling, 3D Printing, Science Fiction Drawing, Creative Writing, Board Games